Phu Vinh Nguyen
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TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Bash, R, HTML/CSS, C#
Full Stack & Web: React, Node.js, Express, Electron, REST APIs, WebSockets, Tailwind CSS
Cloud & DevOps: Kubernetes, Docker, Terraform, GitOps (ArgoCD), AWS (S3), GCP (BigQuery, GCS), Git/GitHub, Linux, RedHat OpenShift, CI/CD, Terraform
Data Engineering: Apache Airflow 3, PySpark, dbt, Kestra, BigQuery, PostgreSQL, MySQL
Observability & BI: Prometheus, Grafana, Grafana Loki, Alloy, Alertmanager, Power BI, Looker Studio

PROFESSIONAL EXPERIENCE

Automation Engineer | Sanofi, Waltham, MA *Jun 2025 – Present*

- Built a support ticketing platform using React for 100+ staff, replacing ad-hoc verbal requests with tracked tickets that route to a team of engineers and trigger email alerts via SMTP.
- Designed a POC WebSocket-based dashboard that streams ticket status and resolution progress live, giving the team real-time visibility into open workload.
- Architected an ingestion pipeline that parses raw instrument metrics and logs into structured records in Prometheus and Loki, powering Grafana dashboards that track instrument performance and robot run status.
- Engineered a C# library for Hamilton VENUS methods to post run data directly to ELNs (Benchling, iLab) through REST APIs.
- Led the observability platform, building and deploying Grafana, Loki, and Prometheus containers on OpenShift and managing the upgrades, secrets, CI/CD deployments, and team onboarding.

Automation Engineer | Foundation Medicine Inc., Boston, MA *Nov 2024 – Apr 2025*

- Designed and delivered ETL pipelines that ingest, validate, and transform sample records into a Sapio LIMS backend using Python and SQL, automating tracking from intake to processing and eliminating a primary source of manual data-entry errors.
- Built a React web app that parses lab instrument configuration files and displays their parameters in a searchable, readable interface, cutting troubleshooting time by 40% for automation engineers and lab operations staff.
- Developed a Python desktop tool (CustomTkinter) that automates renaming, organizing, and updating instrument and sample files to match naming standards, eliminating repetitive manual work.

Automation Engineer (contractor) | Sanofi Pasteur, Waltham, MA *Oct 2022 – Nov 2024*

- Engineered a custom Python toolkit for automated parsing, transformation, and report generation across high-throughput instrument outputs, standardizing data formats across four teams.
- Developed and validated Hamilton VENUS liquid handler methods for high-throughput workflows, automating sample preparation and plate handling to reduce manual pipetting and improve run consistency.

Senior Research Associate | Dermbiont Inc., Boston, MA *Oct 2021 – Sep 2022*

- Operated production data pipelines processing high-throughput genomic datasets across Linux and HPC environments and created containerized QC pipelines alongside software engineers with automated data-quality checks that reduced processing errors 25%.

PROJECTS

E-Commerce Analytics Pipeline | *Airflow 3, PySpark, dbt, GCS, BigQuery, Docker*

- Engineered a DAG Airflow pipeline where a PySpark ingestion layer cleans and enriches 25,000+ records into Parquet on GCS, then triggers a dbt layer via ExternalTaskSensor to materialize revenue and customer KPIs in BigQuery.

Self-hosted Kubernetes Homelab & Observability Stack | *Kubernetes, Prometheus, Grafana, Loki, Alloy, ArgoCD*

- Provisioned a 3-node Kubernetes cluster with kubeadm, hand-configuring control plane, CNI networking, and persistent storage to run PostgreSQL, dedicated game servers, and network services alongside the observability stack.
- Instrumented cluster and workloads with Prometheus and Alloy, authoring custom Grafana dashboards and Alertmanager rules that page a Discord webhook on user-visible symptoms such as service downtime and high memory usage.
  Implemented GitOps-based service management with ArgoCD, versioning all manifests in Git to enable continuous reconciliation and single-command rollback to any prior version.
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EDUCATION

Northeastern University, Boston, MA — Graduate Certificate, Bioinformatics (Software & Data Mgmt) *May 2025*
Uniwersytet Jagielloński, Krakow, Poland — M.S., Molecular Biotechnology *Jul 2021*
University of California, Irvine, Irvine, CA — B.S., Pharmaceutical Sciences *Sep 2019*